

The Linear Regression Model

Lecture 1

The plan of this lecture is to offer a review of regression analysis for the conditional mean function $E(y|x_j, x_{-j})$. We will introduce the basic regression model and then concentrate in the simple case of a bivariate regression. At the end of the lecture, we discuss statistical properties of the Ordinary Least Squares (OLS) estimator.

1 Multivariate Regression Analysis

We start assuming a linear model with p independent variables,

$$\underbrace{y}_{1 \times 1} = \underbrace{x'}_{1 \times p} \underbrace{\beta}_{p \times 1} + \underbrace{u}_{1 \times 1} \quad (1)$$

where y is the response or dependent variable and the β 's are the coefficients associated with the independent variables x 's. At the core of econometrics, there is the concept of causal effects or causal relationships. Obviously, we are not referring to the notion of correlation between x_j and y since $\text{Corr}(x_j, y) = \text{Corr}(y, x_j)$.

We define x_j to be an independent variable, y to be a dependent variable and we might have other (independent) variables we keep at a constant level, x_{-j} . To investigate causal relationships we employ models and the classical model of interest is the conditional expected value of the dependent variable conditional on the independent variables,

$$E(y|x_j, x_{-j}) = E(y|x) = \int y f_{Y|X=x} dx.$$

We can also define:

$$\frac{\partial E(y|x_j, x_{-j})}{\partial x_j} := \text{partial effect}$$

obtained by keeping x_{-j} equal to a constant. Note that this is well defined if x_j is a continuous variable.

To estimate causal effects, a crucial assumption is that all observable factors (the independent variables $x_1, x_2, \dots, x_p$) that affect the dependent variable y are uncorrelated with

the error term.

$$E(u|x_1, x_2, \dots, x_p) = 0. \quad (2)$$

If condition (2) is satisfied, we say that the variables $\{x_1, x_2, \dots, x_p\}$ are strictly exogenous. As explained below, the implication is that the error term has mean equal to zero.

$$E(u|x) = E(u) = 0.$$

To summarize, the first two assumptions of the classical regression model are:

A1 (linearity) The model is linear as in (1)

A2 (strict exogeneity) We assume that $E(u|x_1, x_2, \dots, x_p) = 0$

We will use later a model with p independent variables as in equation (1). Let us now concentrate in one particular case assuming that we have one independent variable, $p = 1$.

2 Conditional Expectations

The simple regression model is,

$$y = \beta_0 + \beta_1 x + u \quad (3)$$

where y is the dependent variable, x is the independent variable, and u is the error term or disturbance. The model suggests that changes in y are induced by changes in x and u . We want to know how much y can be explained in terms of x .

Example 1. Suppose we want to know the effect of education on wages and wages and education are linearly related. Then, we can write

$$wages = \beta_0 + \beta_1 educ + u \quad (4)$$

In this example, we assume that the relation between the response variable and the independent variable is *linear in the parameters* β_0 and β_1 . It may be the case that the model is

$$\log(wages) = \beta_0 + \beta_1 educ + \beta_2 educ^2 + u,$$

which is *non linear in variables*.

We can define the population regression function (PRF) or, in this case, the population regression Line (PRL) of equation (3) as,

$$E(y|x) = E(\beta_0 + \beta_1 x + u|x) = \beta_0 + \beta_1 x + E(u|x) = \beta_0 + \beta_1 x, \quad (5)$$

by Assumptions A1 and A2. It says that the expected value of the dependent variable for a given value of the independent variables is equal to the parameter β_0 plus β_1 times the independent variable. In the case of the PRL with one slope parameter, the model contains two unknown parameters β_0 (the intercept) and β_1 (the slope).

INSERT FIGURE

Our interest lies on estimating the PRF, which implies estimating the two unknowns, i.e. the intercept and the slope.

We now discuss the implications of the previous conditions A1 and A2. We can write the model as:

$$y = E(y|x) + u \quad (6)$$

$$E(u|x) = 0. \quad (7)$$

Equation (7) implies:

$$E(u) = 0 \quad (8)$$

$$Cov(u, x_j) = 0, \text{ for } j = 1, \dots, p, \quad (9)$$

$$Cov(u, h(x_j)) = 0, \text{ for } j = 1, \dots, p, \quad (10)$$

where Cov denotes covariance and $h(\cdot)$ is a known function (e.g. $h(x_1) = x_1^2$, $h(x_1, x_2) = x_1 x_2^2$, $h(x_2) = \exp(x_2 + x_2^2)$, etc.). To see these results, first note that by definition $u := y - E(y|x)$. Then,

$$E(u) = E(y - E(y|x)) = E(y) - E(E(y|x)) = E(y) - E(y) = 0. \quad (11)$$

To show that also implies independence between the error term and covariates, we need to show that

$$E(ux) = E(E(ux|x)) = E(xE(u|x)) = E(x \times 0) = 0. \quad (12)$$

Therefore, conditions (11) and (12) imply, for $j = 1, \dots, p$,

$$Cov(u, x_j) = E(xu) - E(x)E(u) = 0. \quad (13)$$

3 Best Linear Predictor

Define a loss function of a generic form, $L(u) = L(y - m(x))$, where $m(x)$ is the predictor in terms of x and u is the prediction error. Naturally, a decision maker tries to solve the following problem:

$$\min_{m \in \mathcal{M}} E\{L(y - m(x))|x\},$$

that is, she would like to choose a predictor of y that minimizes the expected loss.

It is convenient to assume a quadratic loss (e.g. $L(u) = u^2$). Therefore the problem may be transformed to minimize the mean square error of $m(x)$ for y ,

$$\begin{aligned} E\{(y - m(x))^2|x\} &= E\{(y - \mu(x) + \mu(x) - m(x))^2|x\} \\ &= V(y|x) + (\mu(x) - m(x))^2 + 2E[(m(x) - \mu(x))(y - \mu(x))|x] \\ &= V(y|x) + (\mu(x) - m(x))^2 \end{aligned}$$

where $V(y|x)$ is the conditional variance of y .

If the first two conditional moments of y exists (e.g., $E(y|x) < \infty$ and $V(y|x) < \infty$), the (unconditional) mean square error is minimized to the conditional variance of y when $m(x)$ is equal to the conditional mean $\mu := E(y|x)$,

$$E(y - m(x))^2 = EV(y|x) + E(m(x) - \mu(x))^2.$$

Suppose now that the functional form of m is known. For instance, consider now that a model linear in parameters:

$$\min_{\beta} E\{(y - x'\beta)^2\} = \min_{\beta} \{Ey^2 - 2E(yx')\beta + \beta'E(xx')\beta\}$$

Taking first order condition, we have

$$-E(xy) + E(xx')\beta = 0,$$

thus,

$$\beta = (E(xx'))^{-1}E(xy)$$

Therefore, the parameter β can be interpreted as the best linear predictor under squared error loss.

Note that for β to exists, we need the $p \times p$ matrix $E(xx')$ to be invertible. (We also need that $E(y^2) < \infty$ and $E(\sum_j x_j^2) < \infty$ (see Hansen 2017, Theorem 2.18.1)). If no redundant variables are included in the vector x , then the design matrix $E(xx')$ is invertible and we can say that β is unique and *identified*. A requirement is that $E(xx')$ is positive definite. Identification in this class of linear models means that we can write the parameter of interest in terms of moment conditions that are functions of observed variables such us y and $x_1, \dots, x_p$.

This is related to another assumption of the linear model. The condition

A3 (full rank design matrix) The rank $(E(x_i x_i')) = p$

This condition rules out perfect collinearity, or in other words, it rules out exact linear

relationships among the independent variables.

4 Partial Effects and Elasticities

Consider that we have the following population regression model,

$$E(y|x) = \beta_0 + \sum_{j=1}^p \beta_j x_j,$$

and we are interested in the marginal effect of the j -th independent variable on the conditional expected value of y ,

$$\frac{\partial E(y|x)}{\partial x_j} = \beta_j.$$

This tells us the increase in the mean value of y when the independent variable is increased by 1 unit, other things equal. This interpretation is equivalent to the notion of *ceteris paribus*: the effect of x on y holding everything constant. Note that if x_i and x_j are related, finding the marginal effect is somewhat more complicated, but the linearity of the regression model has a simple and sometimes useful implication for empirical analysis.

Consider the following model with $p = 3$ and $x_3 = x_1 \times x_2$:

$$E(y|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 (x_1 \times x_2).$$

where $\mathbf{x} = (x_1, x_2, x_3)'$. The partial effect of x_1 on the conditional expected value of the dependent variable is:

$$\frac{\partial E(y|\mathbf{x})}{\partial x_1} = \beta_1 + \beta_3 x_2,$$

and,

$$\frac{\partial E(y|\mathbf{x})}{\partial x_2} = \beta_2 + \beta_3 x_1.$$

Sometimes we are interested in the elasticity of y with respect to x_j :

$$\frac{\partial y}{\partial x_j} \frac{x_j}{y}$$

Our partial elasticity is,

$$\frac{\partial E(y|\mathbf{x})}{\partial x_j} \frac{x_j}{E(y|\mathbf{x})} = \frac{\partial \log(E(y|\mathbf{x}))}{\partial \log(x_j)}$$

For instance, the partial elasticity with respect to x_1 in our previous example is:

$$\frac{(\beta_2 + \beta_3 x_2)x_1}{\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 \times x_2}$$

When u and x are not correlated, as it is the case in the previous models, we can have an alternative definition of elasticity that it is more convenient:

$$\frac{\partial E(\log(y)|\mathbf{x})}{\partial \log(x_j)}$$

Note that $E(y|x) = \exp(\mathbf{x}'\boldsymbol{\beta})$ implies,

$$\log(E(y|\mathbf{x})) = \mathbf{x}'\boldsymbol{\beta}.$$

If $y = \exp(\mathbf{x}'\boldsymbol{\beta} + u)$, then $\log(y) = \mathbf{x}'\boldsymbol{\beta} + u$, and by previous assumptions $E(\log(y)) = \mathbf{x}'\boldsymbol{\beta}$ because $E(u|\mathbf{x}) = 0$.

Therefore, the parameter β_j should be interpreted as an elasticity if the model is in a ‘log-log’ form,

$$\log(y) = \beta_0 + \beta_1 \log(x_1) + \beta_2 \log(x_2) + u$$

or as a semielasticity if the model is a log-linear model:

$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

which implies that the partial effect is

$$\frac{\partial E(\log(y)|\mathbf{x})}{\partial x_j}.$$

Lastly, we can introduce the concept of average partial effects (for details, see Blundell and Powell (2003)). Suppose we have the following model,

$$E(y|\mathbf{x}) = g(\mathbf{x}, z),$$

where $\mathbf{x}$ is an observed variable and z is a latent variable. The partial effect is:

$$\frac{\partial E(y|\mathbf{x}, z)}{\partial x_j} = \frac{\partial g(\mathbf{x}, z)}{\partial x_j}$$

Note that the partial effect depends on a latent variable which can take different values from

an underlying distribution. The average partial effect at $\mathbf{x} = \mathbf{x}_0$ is,

$$\delta_j(\mathbf{x}_0) = E\left(\frac{\partial g(\mathbf{x}_0, z)}{\partial x_j}\right) = \int \frac{\partial g(\mathbf{x}_0, z)}{\partial x_j} f(z) dz$$

Can we estimate the average partial effect if we do not observe z ? We essentially have two important cases. If $\mathbf{x}$ and z are independent, the average partial effect is simply the partial effect with respect to x_j :

$$\frac{\partial E(y|\mathbf{x})}{\partial x_j}.$$

Alternatively, if we want to model unobserved heterogeneity, we can find $E(y|\mathbf{x})$ by integrating out over the distribution of the latent variable z .

Example 2. Suppose,

$$y = \beta_0 + \beta_1 x_1 + \gamma_1 x_1 z_1 + u$$

where $E(u|x_1, z_1) = 0$. The conditional mean model is

$$E(y|x, z) = \beta_0 + \beta_1 x_1 + \gamma_1 x_1 z_1$$

and,

$$\frac{\partial E(y|x, z)}{\partial x_j} = \beta_1 + \gamma_1 z_1.$$

If $E(z_1) = E(z_1|x_1) = 0$, then the average partial effect is,

$$E(\beta_1 + \gamma_1 z_1) = \beta_1 = \frac{\partial E(y|x)}{\partial x_j}.$$

If $\mathbf{x}$ and z are not independent, but they are conditional independent on a proxy variable $\mathbf{w}$, it is possible to show that the average partial effect is:

$$\delta_j(\mathbf{x}_0) = E_z\left(\frac{\partial E(y|\mathbf{x}_0, z)}{\partial x_j}\right) = E_z\left(\frac{\partial E(y|\mathbf{x}_0, \mathbf{w})}{\partial x_j}\right).$$

5 Estimation

Consider a random sample $\{y_i, x_i\}$ of size n from population. The pair of data points (y_i, x_i) represent the i^{th} observation. The size of the sample n denotes the total number of observations. Since each data point comes from model (1),

$$y_i = \beta_0 + \beta_1 x_i + u_i \tag{14}$$

where $i = 1, \dots, n$. Our goal is to estimate the parameters (β_0, β_1) .

5.1 Method of Moments

We first review an estimation method called Method of Moments using the following example.

Example 3. Suppose you want to estimate the first moment of a random variable. The method of moments (MM) can be used to estimate the population mean. Consider $\{y_i\}$ to be an independently and identically distributed (i.i.d.) random variable drawn from a distribution whose first moment exists (e.g., y is not distributed as Cauchy). In the population,

$$E(y - \mu) = 0$$

so we can use the corresponding sample moment condition

$$\frac{1}{n} \sum_{i=1}^n (y_i - \mu) = 0 \Rightarrow \hat{\mu} = \bar{y}.$$

To obtain the MM estimator for the linear model, recall the moment condition,

$$E(x_i u_i) = E(E(x_i u_i | x)) = E(x E(u_i | x)) = E(x \times 0) = 0,$$

or alternatively,

$$E(x_i(y_i - x_i' \beta)) = 0,$$

where $x_i = (1, x_{i1})'$ and $\beta = (\beta_0, \beta_1)'$. The MM estimator could be obtained as the solution of the corresponding sample moment condition,

$$\frac{1}{n} \sum_{i=1}^n (x_i(y_i - x_i' \beta)) = 0,$$

yielding,

$$\hat{\beta} = \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \sum_{i=1}^n x_i y_i.$$

We now turn to some basic linear regression models.

5.2 Ordinary Least Squares Estimation (OLS):

Our model's assumptions were

$$\begin{aligned} E(u) &= E(y - \beta_0 - \beta_1 x) = 0 \\ \text{Cov}(x, u) &= E(xu) - E(x)E(u) = E(xu) = E(x(y - \beta_0 - \beta_1 x)) = 0 \end{aligned}$$

These two equations are restrictions imposed to the data about the joint probability distribution of x and y . There are two unknowns and two equations, which suggests that we can solve for β_0 and β_1 . The sample counterpart of these equations are

$$\frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (15)$$

$$\frac{1}{n} \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (16)$$

The Ordinary Least Squares Estimator is the argument that minimizes the following criterion function:

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2, \quad (17)$$

with respect to the unknowns β_0 and β_1 . It is easy to see that the normal equations corresponding to this problem are identical to the two previous equations (15) and (16). Thus, the OLS estimator is a MM estimator in this case.

From equation (15), we obtain

$$\bar{y} - \beta_0 - \beta_1 \bar{x} = 0 \Rightarrow \hat{\beta}_0 = \bar{y} - \beta_1 \bar{x} \quad (18)$$

where $\bar{y} = n^{-1} \sum_{i=1}^n y_i$. Thus, if we know the ordinary least square estimator of the slope β_1 , we can obtain an estimate of the intercept. How can we obtain $\hat{\beta}_1$? Plug the last result in equation (16) to obtain,

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n x_i (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \beta_1 x_i) &= 0 \\ \frac{1}{n} \sum_{i=1}^n x_i ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})) &= 0 \\ \sum_{i=1}^n x_i (y_i - \bar{y}) &= \hat{\beta}_1 \sum_{i=1}^n x_i (x_i - \bar{x}) \end{aligned}$$

It can be shown that the last equation is equal to

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})^2$$

which implies that the estimator for the parameter β_1 is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (19)$$

Therefore, $\hat{\beta}_0, \hat{\beta}_1$ are the ordinary least squares (OLS) estimators for the intercept and slope parameters of the simple linear regression model.

5.3 Arithmetic Properties of OLS

It is useful to briefly discuss properties of the OLS estimators. Consider the following 4 properties:

1. The sum of the residuals $\hat{u}_i$ is zero since $\hat{\beta}_0$ is chosen such that

$$\sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = \sum_{i=1}^n \hat{u}_i = 0$$

2. The sample covariance between the independent variable and the residuals are zero.
To see this, it is crucial to find that

$$\begin{aligned} \sum_{i=1}^n x_i \hat{u}_i &= \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = \sum_{i=1}^n x_i (y_i - x_i' \hat{\beta}) \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i x_i' \hat{\beta} \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i x_i' \left(\sum_{i=1}^n x_i x_i' \right)^{-1} \sum_{i=1}^n x_i y_i \\ &= \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i y_i = 0. \end{aligned}$$

3. The sample average of the fitted values is equal to sample average of the observations.

$$\begin{aligned} y_i &= \hat{y}_i + \hat{u}_i \\ \sum y_i &= \sum \hat{y}_i + \sum \hat{u}_i \\ \frac{1}{n} \sum y_i &= \frac{1}{n} \sum \hat{y}_i \Rightarrow \bar{y} = \bar{\hat{y}} \end{aligned}$$

4. The point $(\bar{x}, \bar{y})$ is always on the regression OLS line.

$$\bar{y} = \bar{\hat{y}} = \frac{1}{n} \sum \hat{y}_i = \frac{1}{n} \sum (\hat{\beta}_0 + \hat{\beta}_1 x_i) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

5.4 Bias of the OLS estimator

Under condition A1 and A2, the OLS estimator is unbiased, that is, the expected value of the OLS estimator is equal to the unknown parameter.

We can show the result as follows. Recall that,

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) y_i \\ &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) (\beta_0 + \beta_1 x_i + u_i) \\ &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \left(\beta_0 \sum_{i=1}^n (x_i - \bar{x}) + \beta_1 \sum_{i=1}^n (x_i - \bar{x}) x_i + \sum_{i=1}^n (x_i - \bar{x}) u_i \right) \\ &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \left(\beta_1 \sum_{i=1}^n (x_i - \bar{x})^2 + \sum_{i=1}^n (x_i - \bar{x}) u_i \right). \end{aligned}$$

Therefore, the OLS estimator for the slope can be written as:

$$\hat{\beta}_1 = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) u_i. \quad (20)$$

Taking expectation,

$$E(\hat{\beta}_1) = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n E((x_i - \bar{x}) u_i)$$

Since $E(u|x) = E(u) = 0$, $\Rightarrow E(\hat{\beta}_1) = \beta_1$. Also, recall that

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} = \beta_0 + \beta_1 \bar{x} - \hat{\beta}_1 \bar{x} \\ \Rightarrow E(\hat{\beta}_0) &= \beta_0 + \beta_1 \bar{x} - E(\hat{\beta}_1) \bar{x} = \beta_0\end{aligned}$$

Therefore, the OLS estimators are unbiased e.g.

$$E(\hat{\beta}_0) = \beta_0; \quad E(\hat{\beta}_1) = \beta_1 \quad (21)$$

5.5 Variance of the OLS estimator

We are going to obtain the variance of the OLS estimators, but we have to state first one of the assumptions of the simple linear regression model:

A4 (homocedasticity) $\text{Var}(u|x) = \sigma^2$

A5 (uncorrelated errors) $\text{Cov}(u_i, u_j) = 0$ for $i, j = 1, \dots, i-1, i+1, \dots, n$

In other words, the variance of the error term is constant (it does not change with x) and the error u_i is independent of u_j for $j \neq i$. Assumptions A4 and A5 together imply spherical errors, or more precisely, spherical error variance.

What is the variance of $\hat{\beta}_1$? We know that

$$\hat{\beta}_1 = \beta_1 + \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) u_i$$

Considering A4 and A5, we obtain the variance as follows:

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var} \left(\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \sum_{i=1}^n (x_i - \bar{x}) u_i \right) \\ &= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \text{Var} \left(\sum_{i=1}^n (x_i - \bar{x}) u_i \right)\end{aligned}$$

$$\begin{aligned}
\text{Var}(\hat{\beta}_1) &= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \sum_{i=1}^n \text{Var}((x_i - \bar{x})u_i) \\
&= \frac{1}{(\sum_{i=1}^n (x_i - \bar{x})^2)^2} \sum_{i=1}^n (x_i - \bar{x})^2 \text{Var}(u_i) \\
\Rightarrow \text{Var}(\hat{\beta}_1) &= \frac{\sigma^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)}
\end{aligned}$$

Similary, to obtain the variance of the intercept, under the stated assumptions, we write,

$$\begin{aligned}
\text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) = \text{Var}\left(\bar{y} - \frac{\sum_i (x_i - \bar{x}) y_i}{\sum_i (x_i - \bar{x})^2} \bar{x}\right) \\
&= \text{Var}\left(\sum_i \left(\frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right) y_i\right) \\
&= \sum_i \left(\frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right)^2 \text{Var}(u_i) \\
&= \sigma^2 \sum_i \left(\frac{1}{n^2} + \left(\frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right)^2 - \frac{2}{n} \frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right) \\
&= \sigma^2 \sum_i \left(\frac{1}{n^2} + \left(\frac{\bar{x}(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2}\right)^2\right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_i (x_i - \bar{x})^2}\right) = \sigma^2 \left(\frac{\sum_i (x_i - \bar{x})^2 + n\bar{x}^2}{n \sum_i (x_i - \bar{x})^2}\right) \\
&= \sigma^2 \left(\frac{\sum_i x_i^2 + n\bar{x}^2 - 2\bar{x} \sum_i x_i + n\bar{x}^2}{n \sum_i (x_i - \bar{x})^2}\right) \\
&= \sigma^2 \left(\frac{\sum_i x_i^2 + n\bar{x}^2 - 2n\bar{x}^2 + n\bar{x}^2}{n \sum_i (x_i - \bar{x})^2}\right) = \sigma^2 \left(\frac{\sum_i x_i^2}{n \sum_i (x_i - \bar{x})^2}\right),
\end{aligned}$$

which leads to the variance of the estimated intercept,

$$\text{Var}(\hat{\beta}_0) = \frac{n^{-1} \sum_{i=1}^n x_i^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)} \sigma^2$$

Although we do not know the parameter σ^2 , we can plugin its estimator $\hat{\sigma}^2$.

Remark 1. The variance of the error term is another unknown parameter of the population, but we can estimate using the analog principle. Recall that u is distributed as a normal

variable with mean zero and variance σ^2 ,

$$E(u) = 0; \quad V(u) = E(u^2) - (E(u))^2 = E(u^2) = \sigma^2,$$

which implies that we can use (from the sample counterpart),

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2$$

as an estimator of σ^2 . Unfortunately, this is a biased estimator, but we can easily correct this to have an unbiased estimator of the variance of the error term,

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2 \quad (22)$$

where $(n-2)$ are the degrees of freedom.

Therefore, the estimator of the variance of the OLS estimator,

$$\text{Var}(\hat{\beta}_0) = \frac{n^{-1} \sum_{i=1}^n x_i^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)} \hat{\sigma}^2; \quad \text{Var}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{(\sum_{i=1}^n (x_i - \bar{x})^2)} \quad (23)$$

Then, we can define the standard error of the slope coefficient as

$$\text{se}(\hat{\beta}_1) = \frac{1}{\sqrt{(\sum_{i=1}^n (x_i - \bar{x})^2)}} \hat{\sigma} \quad (24)$$

6 Multivariate Linear Models

So far, we studied the a simple regression model, which includes only one covariate. It makes sense to extend the previous model to a model with p independent variables,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + u = \beta_0 + \sum_{j=1}^p \beta_j x_j + u \quad (25)$$

where β_0 is the intercept, β_1 is the coefficient associated with the independent variable x_1 , β_2 with the variable x_2 , and β_p with the variable x_p . The random variable u is the error term, which contains other factors than $x_1, x_2, \dots, x_p$.

The crucial assumption is that all observable factors (the independent variables $x_1, x_2, \dots, x_p$) that affect the dependent variable y are uncorrelated with the error term: $E(u|x_1, x_2, \dots, x_p) = 0$.

6.1 Method of Moments

The basic idea here is to apply the restrictions in the population to the data. For instance, recall that we have the assumptions

$$E(u_i) = 0$$

$$E(x_{i1}, u_i) = 0$$

$$E(x_{ip}, u_i) = 0$$

Therefore, we should find the coefficient estimates that solve the following system of equations,

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{i1} \hat{u}_i = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{ip} \hat{u}_i = 0$$

Again, we have $(p+1)$ equations and $(p+1)$ unknowns. The solutions are called the method of moment estimators for the regression parameters $\beta_0, \dots, \beta_p$.

For instance, consider the model with two independent variables. The conditions will be

$$E(u_i) = 0$$

$$E(x_{i1}u_i) = 0$$

$$E(x_{i2}u_i) = 0$$

consequently,

$$\frac{1}{n} \sum_{i=1}^n \hat{u}_i = 0 \Rightarrow \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}) = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{i1} \hat{u}_i = 0 \Rightarrow \sum_{i=1}^n x_{i1} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}) = 0$$

$$\frac{1}{n} \sum_{i=1}^n x_{i2} \hat{u}_i = 0 \Rightarrow \sum_{i=1}^n x_{i2} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \hat{\beta}_2 x_{i2}) = 0$$

As we will see below, the method of moments gives estimates coefficients that are identical to the least squares estimates.

6.2 OLS

The least squares procedure is similar than the one discussed before. It computes the estimated coefficients as minimizers of the sum of residuals squared,

$$S = \sum_{i=1}^n \hat{u}_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The difference is now that

$$y_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + u_i$$

instead of the sample $y_i = \beta_0 + \beta_1 x_i$. For instance, the predicted value of y_i is,

$$\hat{y}_i = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_{ij},$$

under the standard assumptions. To obtain the solution of this problem, we use partial derivatives to develop a set of $(p+1)$ simultaneous normal equations,

$$\begin{aligned} \frac{\partial S}{\partial \hat{\beta}_0} &= 0 \\ \frac{\partial S}{\partial \hat{\beta}_1} &= 0 \\ &\vdots \\ \frac{\partial S}{\partial \hat{\beta}_p} &= 0 \end{aligned}$$

which has $(p+1)$ unknowns $\hat{\beta}_0, \dots, \hat{\beta}_p$. For simplicity, we focus on the problem of minimizing

$$S = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2})^2,$$

with partial derivatives

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n x_{i1} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

$$\frac{\partial S}{\partial \beta_2} = -2 \sum_{i=1}^n x_{i2}(y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2}) = 0$$

we can write these equations as

$$\begin{aligned} n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_{i1} + \hat{\beta}_2 \sum_{i=1}^n x_{i2} &= \sum_{i=1}^n y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_{i1} + \hat{\beta}_1 \sum_{i=1}^n x_{i1}^2 + \hat{\beta}_2 \sum_{i=1}^n x_{i1}x_{i2} &= \sum_{i=1}^n x_{i1}y_i \\ \hat{\beta}_0 \sum_{i=1}^n x_{i2} + \hat{\beta}_1 \sum_{i=1}^n x_{i1}x_{i2} + \hat{\beta}_2 \sum_{i=1}^n x_{i2}^2 &= \sum_{i=1}^n x_{i2}y_i \end{aligned}$$

It can be shown that the estimate for the intercept coefficient β_0 is,

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2.$$

It is somewhat more tedious to find that

$$\begin{aligned} \hat{\beta}_1 &= \frac{S_y(r_{x_1,y} - r_{x_1x_2}r_{x_2y})}{S_{x_1}(1 - r_{x_1x_2}^2)} \\ \hat{\beta}_2 &= \frac{S_y(r_{x_2,y} - r_{x_1x_2}r_{x_1y})}{S_{x_2}(1 - r_{x_1x_2}^2)} \end{aligned}$$

Where r_{x_jy} is the sample correlation between x_j and y , $r_{x_1x_2}$ is the sample correlation between x_1 and x_2 , S_{x_j} is the sample standard deviation for x_j , and S_y is the sample standard deviation for y .

What we can learn from these estimators is that in the multiple regression model the estimate of the regression coefficient β_1 depends on the correlation between x_1 and x_2 , and r_{x_2} and y . If there is no correlation between x_1 and x_2 , (e.g., $r_{x_1,x_2} = 0$), the estimate for the coefficient β is,

$$\begin{aligned} \hat{\beta}_1 &= \frac{S_y r_{x_1y}}{S_{x_1}} \\ &= \frac{\sum_{i=1}^n (y_i - \bar{y})(x_{i1} - \bar{x})}{\sum_{i=1}^n (x_{i1} - \bar{x})^2} \end{aligned}$$

If $r_{x_1,x_2} = 1$, the estimate of the coefficient β_1 will not be defined. We could obtain the same conclusions on $\hat{\beta}_2$. In empirical analysis, it is always important to explore the correlation between the independent variables.

7 Analysis-of-variance and R^2

Consider,

$$\begin{aligned} y_i &= \hat{y}_i + \hat{u}_i \\ y_i - \bar{y} &= (\hat{y}_i - \bar{y}) + \hat{u}_i \\ \sum (y_i - \bar{y})^2 &= \sum (\hat{y}_i - \bar{y})^2 + 2 \sum \hat{u}_i (\hat{y}_i - \bar{y}) + \sum \hat{u}_i^2 \end{aligned}$$

Note that using the arithmetic properties of the OLS estimators,

$$\sum \hat{u}_i (\hat{y}_i - \bar{y}) = \sum \hat{u}_i (\hat{\beta}_1 (x_i - \bar{x})) = \hat{\beta}_1 (\sum \hat{u}_i x_i - \bar{x} \sum \hat{u}_i) = 0$$

Therefore,

$$\sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum \hat{u}_i^2. \quad (26)$$

If we divide the expression by the right hand side variable,

$$1 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} + \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2}$$

which indicates a proportion that it is explained by the linear estimated model (first term) and that it is not explained by the linear model (second term). The R^2 of a regression is defined as

$$R^2 := \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2}$$

Note that a small sum of residuals squared means that the model explains very well, and sum of residuals squared large means that the model does not produce a good fit to the data. Therefore, R^2 close to 1 is telling us that the model produces a good fit to the data, and a R^2 close to 0 says that the model does not produce a good fit to the data. Note, also, that $0 \leq R^2 \leq 1$ since $0 < \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2} < 1$.

An alternative statistic is the adjusted R^2 , which is defined as

$$\bar{R}^2 := 1 - \frac{n-1}{n-p} \frac{\sum \hat{u}_i^2}{\sum (y_i - \bar{y})^2},$$

and provides a trade-off between goodness-of-fit and parsimony. The goodness-of-fit is measured by the R^2 and the parsimony of the model is measured by the degrees of freedom, $n - p$. Note that the $\bar{R}^2$ may not be well defined and can be negative too.

It is immediately apparent that

$$1 - \bar{R}^2 := \frac{n-1}{n-p} (1 - R^2) .$$